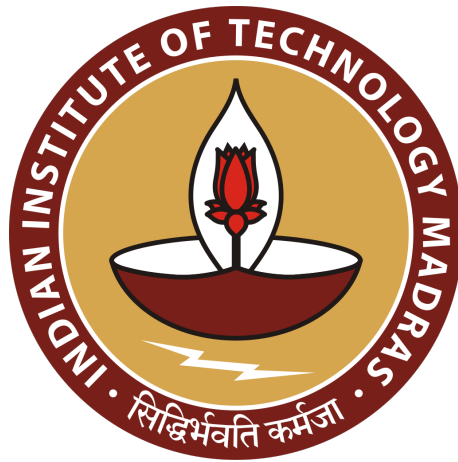


Lab Assignment - 7

Protein Sequence Analysis



BT3040 - Bioinformatics

Even Semester 2026

Course Instructor: Professor Michael Gromiha

K.Varunkumar | BE23B016

Department of Biotechnology

Indian Institute of Technology, Madras

Question 1

The code to compute the Amino Acid composition of the given sequences is given:

```
# Question 1
def calc_composition(seq):

    comp = {}
    AAs = "ACDEFGHIKLMNPQRSTVWY"

    for AA in AAs:
        comp[AA] = False

    for residue in seq:
        if comp.get(residue) is False:
            comp[residue] = 1
        else:
            comp[residue] += 1

    for residue in comp.keys():
        comp[residue] = (comp[residue] * 100) / len(seq)

    return(comp)
```

The output for the above code is shown below:

```
# For sequence 1
The residue A has a composition of: 17.91 %
The residue C has a composition of: 1.49 %
The residue D has a composition of: 2.99 %
The residue E has a composition of: 2.99 %
The residue F has a composition of: 1.49 %
The residue G has a composition of: 5.97 %
The residue H has a composition of: 4.48 %
The residue I has a composition of: 1.49 %
The residue K has a composition of: 7.46 %
The residue L has a composition of: 8.96 %
The residue M has a composition of: 0.0 %
The residue N has a composition of: 1.49 %
The residue P has a composition of: 7.46 %
The residue Q has a composition of: 1.49 %
The residue R has a composition of: 5.97 %
The residue S has a composition of: 5.97 %
The residue T has a composition of: 10.45 %
The residue V has a composition of: 1.49 %
The residue W has a composition of: 5.97 %
The residue Y has a composition of: 4.48 %
```

```
# For sequence 2
The residue A has a composition of: 19.12 %
The residue C has a composition of: 0.0 %
```

The residue D has a composition of: 2.94 %
The residue E has a composition of: 1.47 %
The residue F has a composition of: 5.88 %
The residue G has a composition of: 14.71 %
The residue H has a composition of: 0.0 %
The residue I has a composition of: 11.76 %
The residue K has a composition of: 1.47 %
The residue L has a composition of: 13.24 %
The residue M has a composition of: 7.35 %
The residue N has a composition of: 0.0 %
The residue P has a composition of: 4.41 %
The residue Q has a composition of: 2.94 %
The residue R has a composition of: 2.94 %
The residue S has a composition of: 0.0 %
The residue T has a composition of: 1.47 %
The residue V has a composition of: 8.82 %
The residue W has a composition of: 0.0 %
The residue Y has a composition of: 1.47 %

For sequence 3

The residue A has a composition of: 10.6 %
The residue C has a composition of: 0.0 %
The residue D has a composition of: 5.96 %
The residue E has a composition of: 3.97 %
The residue F has a composition of: 5.3 %
The residue G has a composition of: 15.23 %
The residue H has a composition of: 1.32 %
The residue I has a composition of: 5.3 %
The residue K has a composition of: 3.97 %
The residue L has a composition of: 5.96 %
The residue M has a composition of: 0.66 %
The residue N has a composition of: 3.31 %
The residue P has a composition of: 2.65 %
The residue Q has a composition of: 3.31 %
The residue R has a composition of: 3.31 %
The residue S has a composition of: 9.93 %
The residue T has a composition of: 4.64 %
The residue V has a composition of: 5.3 %
The residue W has a composition of: 3.97 %
The residue Y has a composition of: 5.3 %

- Sequence 1, has A, T, L, P, K as the majority of it's sequence
- Sequence 2, has A, G, L, I, V as the majority of it's sequence, all hydrophobic residues.
- Sequence 3, has G, A, S, L, D as the majority of it's sequence, indicating it might have disulphide bonds in it's tertiary structure.
- In general all 3 sequences appear to be rich in hydrophobic residues, with the order being Sequence 2 > Sequence 3 > Sequence 1 (This is validated in the Question 5).

Question 2

The code for calculating the molecular weights of the given sequences is given below:

```
# Question 2
def Sequence_Weight(seq):

    AA_W = {
        'A': 85,    # Ala
        'C': 115,   # Cys
        'D': 130,   # Asp
        'E': 145,   # Glu
        'F': 160,   # Phe
        'G': 70,    # Gly
        'W': 200,   # Trp
        'H': 150,   # His
        'I': 125,   # Ile
        'K': 145,   # Lys
        'L': 125,   # Leu
        'M': 143,   # Met
        'N': 130,   # Asn
        'Y': 175,   # Tyr
        'P': 110,   # Pro
        'Q': 140,   # Gln
        'R': 170,   # Arg
        'S': 100,   # Ser
        'T': 115,   # Thr
        'V': 110    # Val
    }

    weight = 0

    for residue in seq:
        weight += AA_W[residue]

    weight -= ((len(seq)-1)*18)

    print(f"The molecular weight of the given sequence is: {weight}")
    print("="*100)

Sequence_Weight(seq1)
Sequence_Weight(seq2)
Sequence_Weight(seq3)
```

The output for the above code is shown below:

```
The molecular weight of the given sequence is: 7127
```

```
=====
```

```
The molecular weight of the given sequence is: 6529
```

```
=====
```

```
The molecular weight of the given sequence is: 15453
```

Question 3

The code for determining which group an AA in a sequence belongs to, and which group the entire sequence belongs to is given below:

```
# Question 3

def similarity(seq):
    Group_data = {
        'A': (8.47, 8.95), # Ala
        'D': (5.97, 5.91), # Asp
        'C': (1.39, 0.47), # Cys
        'E': (6.32, 4.78), # Glu
        'T': (5.79, 6.54), # Thr
        'F': (3.91, 3.68), # Phe
        'G': (7.82, 8.54), # Gly
        'H': (2.26, 1.25), # His
        'I': (5.71, 4.77), # Ile
        'V': (7.02, 6.76), # Val
        'K': (5.76, 4.93), # Lys
        'L': (8.48, 8.78), # Leu
        'M': (2.21, 1.56), # Met
        'N': (4.54, 5.74), # Asn
        'W': (1.44, 1.24), # Trp
        'P': (4.63, 3.74), # Pro
        'Q': (3.82, 4.75), # Gln
        'R': (4.93, 5.24), # Arg
        'S': (5.94, 8.05), # Ser
        'Y': (3.58, 4.13)  # Tyr
    }

    comp = calc_composition(seq)
    scores_1, scores_2 = [], []

    sum_sc_1, sum_sc_2 = 0, 0

    AAs = "ACDEFGHIKLMNPQRSTVWY"
    for AA in AAs:
        scores_1 = abs(comp[AA]-Group_data[AA][0])
        scores_2 = abs(comp[AA]-Group_data[AA][1])
        sum_sc_1 += scores_1
        sum_sc_2 += scores_2

    if scores_1 > scores_2:
        print(f"Given sequence's {AA} is similar to Group A")
    elif scores_2 > scores_1:
        print(f"Given sequence's {AA} is similar to Group B")
    else:
        print(f"Given sequence's {AA} is similar to Group A and Group B")

# print(sum_sc_1,sum_sc_2)
if sum_sc_1 < sum_sc_2:
```

```
    print(f"Given sequence is similar to Group A")
elif sum_sc_1 > sum_sc_2:
    print(f"Given sequence is similar to Group B")
else:
    print(f"Given sequence is similar to both Groups")

print("="*100)

similarity(seq1)
similarity(seq2)
similarity(seq3)
```

The output for the above code is shown below:

```
# Sequence 1
Given sequence's A is similar to Group A
Given sequence's C is similar to Group B
Given sequence's D is similar to Group A
Given sequence's E is similar to Group A
Given sequence's F is similar to Group A
Given sequence's G is similar to Group B
Given sequence's H is similar to Group B
Given sequence's I is similar to Group A
Given sequence's K is similar to Group B
Given sequence's L is similar to Group A
Given sequence's M is similar to Group A
Given sequence's N is similar to Group B
Given sequence's P is similar to Group B
Given sequence's Q is similar to Group B
Given sequence's R is similar to Group A
Given sequence's S is similar to Group B
Given sequence's T is similar to Group A
Given sequence's V is similar to Group A
Given sequence's W is similar to Group B
Given sequence's Y is similar to Group A
Given sequence is similar to Group A
```

```
# Sequence 2
Given sequence's A is similar to Group A
Given sequence's C is similar to Group A
Given sequence's D is similar to Group A
Given sequence's E is similar to Group A
Given sequence's F is similar to Group B
Given sequence's G is similar to Group A
Given sequence's H is similar to Group A
Given sequence's I is similar to Group B
Given sequence's K is similar to Group A
Given sequence's L is similar to Group A
Given sequence's M is similar to Group B
Given sequence's N is similar to Group B
Given sequence's P is similar to Group B
```

```
Given sequence's Q is similar to Group B
Given sequence's R is similar to Group B
Given sequence's S is similar to Group B
Given sequence's T is similar to Group B
Given sequence's V is similar to Group B
Given sequence's W is similar to Group A
Given sequence's Y is similar to Group B
Given sequence is similar to Group A
```

```
# Sequence 3
Given sequence's A is similar to Group A
Given sequence's C is similar to Group A
Given sequence's D is similar to Group B
Given sequence's E is similar to Group A
Given sequence's F is similar to Group B
Given sequence's G is similar to Group A
Given sequence's H is similar to Group A
Given sequence's I is similar to Group B
Given sequence's K is similar to Group A
Given sequence's L is similar to Group B
Given sequence's M is similar to Group A
Given sequence's N is similar to Group B
Given sequence's P is similar to Group A
Given sequence's Q is similar to Group B
Given sequence's R is similar to Group B
Given sequence's S is similar to Group A
Given sequence's T is similar to Group B
Given sequence's V is similar to Group A
Given sequence's W is similar to Group B
Given sequence's Y is similar to Group A
Given sequence is similar to Group B
```

- Sequence 1 is thus similar to Group A
- Sequence 2 is thus similar to Group A
- Sequence 3 is thus similar to Group B

Question 4

The code for calculating the residue pair preference for the given sequences is given below:

```
# Question 4
def count_di(seq):
    Scores_1 = np.zeros((20, 20))
    Scores_2 = np.zeros((20, 20))
    Scores_3 = np.zeros((20, 20))
    Score_list_1, Score_list_2, Score_list_3 = [], [], []

    AAs = "ACDEFGHIKLMNPQRSTVWY"
    di_nuc = {}
```

```

for n in range(len(seq)-1):
    di = seq[n:n+2]
    if di_nuc.get(di) is None:
        di_nuc[di] = 1
    else:
        di_nuc[di] += 1

comp = {}
for AA in AAs:
    comp[AA] = 0

for residue in seq:
    comp[residue] += 1

for n in range(20):
    for m in range(20):
        if di_nuc.get(AAs[n]+AAs[m]) is not None:
            Scores_1[n][m] = (di_nuc[AAs[n]+AAs[m]]*100)/(comp[AAs[n]] + comp[AAs[m]])
            Score_list_1.append((Scores_1[n][m], AAs[n]+AAs[m]))

            Scores_2[n][m] = (di_nuc[AAs[n]+AAs[m]]*100)/(len(seq)-1)
            Score_list_2.append((Scores_2[n][m], AAs[n]+AAs[m]))

            Scores_3[n][m] = (di_nuc[AAs[n]+AAs[m]]*100)/(comp[AAs[n]] * comp[AAs[m]])
            Score_list_3.append((Scores_3[n][m], AAs[n]+AAs[m]))

pd.set_option('display.max_columns', None)
pd.set_option('display.width', None)
pd.set_option('display.expand_frame_repr', False)

aa = list("ACDEFGHIKLMNPQRSTVWY")
df1 = pd.DataFrame(Scores_1, index=aa, columns=aa)
print(df1.round(2))
Score_list_1.sort(reverse = True)
for k in range(min(10,len(Score_list_1))):
    print(f"{k+1}) The residue pair: {Score_list_1[k][1]}, with an occurrence of {round(Score_list_1[k][0],3)} %")

df2 = pd.DataFrame(Scores_2, index=aa, columns=aa)
print(df2.round(2))
Score_list_2.sort(reverse = True)
for k in range(min(10,len(Score_list_2))):
    print(f"{k+1}) The residue pair: {Score_list_2[k][1]}, with an occurrence of {round(Score_list_2[k][0], 3)} %")

df3 = pd.DataFrame(Scores_3, index=aa, columns=aa)
print(df3.round(2))
Score_list_3.sort(reverse = True)
for k in range(min(10,len(Score_list_3))):
    print(f"{k+1}) The residue pair: {Score_list_3[k][1]}, with an occurrence of {round(Score_list_3[k][0], 3)} %")

```


For Sequence 1

	A	C	D	E	F	G	H	I	K	L	M	N	P	Q	R	S	T	V	W	Y
A	12.50	0.0	0.0	0.00	0.0	6.25	0.00	0.0	0.00	11.11	0.0	0.0	0.00	0.00	0.00	6.25	15.79	0.00	6.25	0.00
C	0.00	0.0	0.0	0.00	50.0	0.00	0.00	0.0	0.00	0.00	0.0	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
D	0.00	0.0	0.0	25.00	0.0	16.67	0.00	0.0	0.00	0.00	0.0	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
E	7.14	0.0	0.0	0.00	0.0	0.00	0.00	0.0	0.00	0.00	0.0	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	20.00
F	0.00	0.0	0.0	0.00	0.0	0.00	0.00	0.0	0.00	0.00	0.0	50.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
G	0.00	20.0	0.0	0.00	0.0	0.00	0.00	20.0	0.00	10.00	0.0	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	14.29
H	13.33	0.0	0.0	0.00	0.0	0.00	16.67	0.0	0.00	0.00	0.0	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
I	0.00	0.0	0.0	0.00	0.0	0.00	0.00	0.0	16.67	0.00	0.0	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
K	5.88	0.0	0.0	0.00	0.0	0.00	0.00	0.0	0.00	0.00	0.0	0.0	10.00	16.67	0.00	11.11	0.00	0.00	11.11	0.00
L	5.56	0.0	12.5	0.00	0.0	0.00	11.11	0.0	9.09	8.33	0.0	0.0	0.00	0.00	0.00	0.00	7.69	0.00	0.00	0.00
M	0.00	0.0	0.0	0.00	0.0	0.00	0.00	0.0	0.00	0.00	0.0	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
N	0.00	0.0	0.0	0.00	0.0	0.00	0.00	0.0	0.00	0.00	0.0	0.0	0.00	0.00	20.00	0.00	0.00	0.00	0.00	0.00
P	0.00	0.0	0.0	14.29	0.0	0.00	0.00	0.0	0.00	0.00	0.0	0.0	0.00	0.00	0.00	11.11	8.33	16.67	11.11	0.00
Q	0.00	0.0	0.0	0.00	0.0	0.00	0.00	0.0	16.67	0.00	0.0	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
R	6.25	0.0	0.0	0.00	0.0	0.00	14.29	0.0	0.00	0.00	0.0	0.0	11.11	0.00	0.00	0.00	0.00	0.00	12.50	0.00
S	0.00	0.0	0.0	0.00	0.0	0.00	0.00	0.0	11.11	10.00	0.0	0.0	0.00	0.00	12.50	0.00	0.00	0.00	0.00	14.29
T	10.53	0.0	0.0	0.00	0.0	9.09	0.00	0.0	8.33	0.00	0.0	0.0	16.67	0.00	9.09	0.00	0.00	0.00	0.00	0.00
V	0.00	0.0	0.0	0.00	0.0	20.00	0.00	0.0	0.00	0.00	0.0	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
W	0.00	0.0	0.0	0.00	0.0	0.00	0.00	0.0	0.00	0.00	0.0	0.0	11.11	0.00	0.00	12.50	18.18	0.00	0.00	0.00
Y	6.67	0.0	20.0	0.00	0.0	0.00	0.00	0.0	0.00	11.11	0.0	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1) The residue pair: FN, with an occurrence of 50.0 %																				
2) The residue pair: CF, with an occurrence of 50.0 %																				
3) The residue pair: DE, with an occurrence of 25.0 %																				
4) The residue pair: YD, with an occurrence of 20.0 %																				
5) The residue pair: VG, with an occurrence of 20.0 %																				
6) The residue pair: NR, with an occurrence of 20.0 %																				
7) The residue pair: GI, with an occurrence of 20.0 %																				
8) The residue pair: GC, with an occurrence of 20.0 %																				
9) The residue pair: EY, with an occurrence of 20.0 %																				
10) The residue pair: WT, with an occurrence of 18.182 %																				

Figure 4.1.1: Score 1: $\left[\frac{N_{ij} \times 100}{N_i + N_j} \right]$

	A	C	D	E	F	G	H	I	K	L	M	N	P	Q	R	S	T	V	W	Y
A	4.55	0.00	0.00	0.00	0.00	1.52	0.00	0.00	0.00	3.03	0.0	0.00	0.00	0.00	0.00	1.52	4.55	0.00	1.52	0.00
C	0.00	0.00	0.00	0.00	1.52	0.00	0.00	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
D	0.00	0.00	0.00	1.52	0.00	1.52	0.00	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
E	1.52	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.52
F	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.0	1.52	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
G	0.00	1.52	0.00	0.00	0.00	0.00	0.00	1.52	0.00	1.52	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.52
H	3.03	0.00	0.00	0.00	0.00	0.00	1.52	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
I	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.52	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
K	1.52	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.00	1.52	1.52	0.00	1.52	0.00	0.00	1.52	0.00
L	1.52	0.00	1.52	0.00	0.00	0.00	1.52	0.00	1.52	1.52	0.0	0.00	0.00	0.00	0.00	0.00	1.52	0.00	0.00	0.00
M	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
N	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.00	0.00	0.00	1.52	0.00	0.00	0.00	0.00	0.00
P	0.00	0.00	0.00	1.52	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.00	1.52	1.52	1.52	1.52	0.00
Q	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.52	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
R	1.52	0.00	0.00	0.00	0.00	0.00	1.52	0.00	0.00	0.00	0.0	0.00	1.52	0.00	0.00	0.00	0.00	0.00	1.52	0.00
S	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.52	1.52	0.0	0.00	0.00	0.00	1.52	0.00	0.00	0.00	0.00	1.52
T	3.03	0.00	0.00	0.00	0.00	1.52	0.00	0.00	1.52	0.00	0.0	0.00	3.03	0.00	1.52	0.00	0.00	0.00	0.00	0.00
V	0.00	0.00	0.00	0.00	0.00	1.52	0.00	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
W	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.00	1.52	0.00	0.00	1.52	3.03	0.00	0.00	0.00
Y	1.52	0.00	1.52	0.00	0.00	0.00	0.00	0.00	0.00	1.52	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1) The residue pair: AT, with an occurrence of 4.545 %																				
2) The residue pair: AA, with an occurrence of 4.545 %																				
3) The residue pair: WT, with an occurrence of 3.03 %																				
4) The residue pair: TP, with an occurrence of 3.03 %																				
5) The residue pair: TA, with an occurrence of 3.03 %																				
6) The residue pair: HA, with an occurrence of 3.03 %																				
7) The residue pair: AL, with an occurrence of 3.03 %																				
8) The residue pair: YL, with an occurrence of 1.515 %																				
9) The residue pair: YD, with an occurrence of 1.515 %																				
10) The residue pair: YA, with an occurrence of 1.515 %																				

Figure 4.1.2: Score 2: $\left[\frac{N_{ij} \times 100}{N-1} \right]$

	A	C	D	E	F	G	H	I	K	L	M	N	P	Q	R	S	T	V	W	Y
A	2.08	0.0	0.00	0.0	0.0	2.08	0.00	0.0	0.00	2.78	0.0	0.0	0.00	0.0	0.00	2.08	3.57	0.0	2.08	0.00
C	0.00	0.0	0.00	0.0	100.0	0.00	0.00	0.0	0.00	0.00	0.0	0.0	0.00	0.0	0.00	0.00	0.00	0.0	0.00	0.00
D	0.00	0.0	0.00	25.0	0.0	12.50	0.00	0.0	0.00	0.00	0.0	0.00	0.00	0.0	0.00	0.00	0.00	0.0	0.00	0.00
E	4.17	0.0	0.00	0.0	0.0	0.00	0.00	0.0	0.00	0.00	0.0	0.0	0.00	0.0	0.00	0.00	0.00	0.0	0.00	16.67
F	0.00	0.0	0.00	0.0	0.0	0.00	0.00	0.0	0.00	0.00	0.0	100.0	0.00	0.0	0.00	0.00	0.00	0.0	0.00	0.00
G	0.00	25.0	0.00	0.0	0.0	0.00	0.00	25.0	0.00	4.17	0.0	0.0	0.00	0.0	0.00	0.00	0.00	0.0	0.00	8.33
H	5.56	0.0	0.00	0.0	0.0	0.00	11.11	0.0	0.00	0.00	0.0	0.0	0.00	0.0	0.00	0.00	0.00	0.0	0.00	0.00
I	0.00	0.0	0.00	0.0	0.0	0.00	0.00	0.0	20.00	0.00	0.0	0.0	0.00	0.0	0.00	0.00	0.00	0.0	0.00	0.00
K	1.67	0.0	0.00	0.0	0.0	0.00	0.00	0.0	0.00	0.00	0.0	0.0	4.00	20.0	0.00	5.00	0.00	0.0	5.00	0.00
L	1.39	0.0	8.33	0.0	0.0	0.00	5.56	0.0	3.33	2.78	0.0	0.0	0.00	0.0	0.00	0.00	2.38	0.0	0.00	0.00
M	0.00	0.0	0.00	0.0	0.0	0.00	0.00	0.0	0.00	0.00	0.0	0.0	0.00	0.0	0.00	0.00	0.00	0.0	0.00	0.00
N	0.00	0.0	0.00	0.0	0.0	0.00	0.00	0.0	0.00	0.00	0.0	0.0	0.00	0.0	25.00	0.00	0.00	0.0	0.00	0.00
P	0.00	0.0	0.00	10.0	0.0	0.00	0.00	0.0	0.00	0.00	0.0	0.0	0.00	0.0	0.00	5.00	2.86	20.0	5.00	0.00
Q	0.00	0.0	0.00	0.0	0.0	0.00	0.00	0.0	20.00	0.00	0.0	0.0	0.00	0.0	0.00	0.00	0.00	0.0	0.00	0.00
R	2.08	0.0	0.00	0.0	0.0	0.00	8.33	0.0	0.00	0.00	0.0	0.0	5.00	0.0	0.00	0.00	0.00	0.0	6.25	0.00
S	0.00	0.0	0.00	0.0	0.0	0.00	0.00	0.0	5.00	4.17	0.0	0.0	0.00	0.0	6.25	0.00	0.00	0.0	0.00	8.33
T	2.38	0.0	0.00	0.0	0.0	3.57	0.00	0.0	2.86	0.00	0.0	0.0	5.71	0.0	3.57	0.00	0.00	0.0	0.00	0.00
V	0.00	0.0	0.00	0.0	0.0	25.00	0.00	0.0	0.00	0.00	0.0	0.0	0.00	0.0	0.00	0.00	0.00	0.0	0.00	0.00
W	0.00	0.0	0.00	0.0	0.0	0.00	0.00	0.0	0.00	0.00	0.0	0.0	5.00	0.0	0.00	6.25	7.14	0.0	0.00	0.00
Y	2.78	0.0	16.67	0.0	0.0	0.00	0.00	0.0	0.00	5.56	0.0	0.0	0.00	0.0	0.00	0.00	0.00	0.0	0.00	0.00
1)	The residue pair: FM, with an occurrence of 100.0 %																			
2)	The residue pair: CF, with an occurrence of 100.0 %																			
3)	The residue pair: VG, with an occurrence of 25.0 %																			
4)	The residue pair: NR, with an occurrence of 25.0 %																			
5)	The residue pair: GI, with an occurrence of 25.0 %																			
6)	The residue pair: GC, with an occurrence of 25.0 %																			
7)	The residue pair: DE, with an occurrence of 25.0 %																			
8)	The residue pair: QK, with an occurrence of 20.0 %																			
9)	The residue pair: PV, with an occurrence of 20.0 %																			
10)	The residue pair: KO, with an occurrence of 20.0 %																			

For Sequence 2

	A	C	D	E	F	G	H	I	K	L	M	N	P	Q	R	S	T	V	W	Y
A	19.23	0.0	0.0	0.0	0.00	0.00	0.0	14.29	0.00	0.00	0.00	0.0	0.00	0.00	6.67	0.0	0.00	15.79	0.0	0.0
C	0.00	0.0	0.0	0.0	0.00	0.00	0.0	0.00	0.00	0.00	0.00	0.0	0.00	0.00	0.0	0.00	0.00	0.00	0.0	0.0
D	6.67	0.0	0.0	0.0	0.00	0.00	0.0	0.00	0.00	9.09	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.0
E	0.00	0.0	0.0	0.0	0.00	0.00	0.0	0.00	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.0
F	5.88	0.0	0.0	0.0	12.50	0.00	0.0	8.33	0.00	7.69	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.0
G	8.70	0.0	0.0	0.0	0.00	5.00	0.0	11.11	9.09	21.05	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.0
H	0.00	0.0	0.0	0.0	0.00	0.00	0.0	0.00	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.0
I	4.76	0.0	0.0	0.0	0.00	16.67	0.0	0.00	0.00	5.88	0.00	0.0	18.18	0.00	0.00	0.00	0.00	7.14	0.0	0.0
K	0.00	0.0	0.0	0.0	20.00	0.00	0.0	0.00	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.0
L	4.55	0.0	0.0	10.0	0.00	10.53	0.0	5.88	0.00	5.56	0.00	0.0	0.00	0.00	9.09	0.0	0.00	6.67	0.0	10.0
M	0.00	0.0	0.0	0.0	11.11	13.33	0.0	7.69	0.00	0.00	10.00	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.0
N	0.00	0.0	0.0	0.0	0.00	0.00	0.0	0.00	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.0
P	0.00	0.0	20.0	0.0	0.00	0.00	0.0	0.00	0.00	8.33	12.50	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.0
Q	0.00	0.0	0.0	0.0	16.67	0.00	0.0	0.00	0.00	0.00	0.00	0.0	20.00	0.00	0.00	0.00	0.00	0.00	0.0	0.0
R	0.00	0.0	0.0	0.0	0.00	0.00	0.0	0.00	0.00	0.00	0.00	0.0	0.00	25.00	0.00	0.0	33.33	0.00	0.0	0.0
S	0.00	0.0	0.0	0.0	0.00	0.00	0.0	0.00	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.0
T	0.00	0.0	0.0	0.0	0.00	0.00	0.0	0.00	0.00	0.00	0.00	0.0	0.00	33.33	0.00	0.0	0.00	0.00	0.0	0.0
V	5.26	0.0	12.5	0.0	0.00	6.25	0.0	0.00	0.00	0.00	27.27	0.0	0.00	0.00	0.00	0.0	0.00	0.00	0.0	0.0
W	0.00	0.0	0.0	0.0	0.00	0.00	0.0	0.00	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.0	0.00	0.00	0.0	0.0
Y	0.00	0.0	0.0	0.0	0.00	0.00	0.0	0.00	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.0	0.00	14.29	0.0	0.0
1) The residue pair: TQ, with an occurrence of 33.333 %																				
2) The residue pair: RT, with an occurrence of 33.333 %																				
3) The residue pair: VM, with an occurrence of 27.273 %																				
4) The residue pair: RQ, with an occurrence of 25.0 %																				
5) The residue pair: GL, with an occurrence of 21.053 %																				
6) The residue pair: QP, with an occurrence of 20.0 %																				
7) The residue pair: PD, with an occurrence of 20.0 %																				
8) The residue pair: KF, with an occurrence of 20.0 %																				
9) The residue pair: AA, with an occurrence of 19.231 %																				
10) The residue pair: IP, with an occurrence of 18.182 %																				

Figure 4.2.1: Score 1: $\left[\frac{N_{ij} \times 100}{N_i + N_j} \right]$

	A	C	D	E	F	G	H	I	K	L	M	N	P	Q	R	S	T	V	W	Y
A	7.46	0.0	0.00	0.00	0.00	0.00	0.0	4.48	0.00	0.00	0.00	0.0	0.00	0.00	1.49	0.0	0.00	4.48	0.0	0.00
C	0.00	0.0	0.00	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.0	0.00	0.00	0.0	0.00
D	1.49	0.0	0.00	0.00	0.00	0.00	0.0	0.00	0.00	1.49	0.00	0.0	0.00	0.00	0.00	0.0	0.00	0.00	0.0	0.00
E	0.00	0.0	0.00	0.00	0.00	0.00	1.49	0.0	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.0	0.00	0.00	0.0	0.00
F	1.49	0.0	0.00	0.00	0.00	1.49	0.00	0.0	1.49	0.00	1.49	0.0	0.00	0.00	0.00	0.0	0.00	0.00	0.0	0.00
G	2.99	0.0	0.00	0.00	0.00	1.49	0.0	2.99	1.49	5.97	0.00	0.0	0.00	0.00	0.00	0.0	0.00	0.00	0.0	0.00
H	0.00	0.0	0.00	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.0	0.00	0.00	0.0	0.00
I	1.49	0.0	0.00	0.00	0.00	4.48	0.0	0.00	0.00	1.49	0.00	0.0	2.99	0.00	0.00	0.0	0.00	1.49	0.0	0.00
K	0.00	0.0	0.00	0.00	1.49	0.00	0.0	0.00	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.0	0.00	0.00	0.0	0.00
L	1.49	0.0	0.00	1.49	0.00	2.99	0.0	1.49	0.00	1.49	0.00	0.0	0.00	0.00	0.00	1.49	0.00	1.49	0.0	1.49
M	0.00	0.0	0.00	0.00	1.49	2.99	0.0	1.49	0.00	0.00	1.49	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.00
N	0.00	0.0	0.00	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.00
P	0.00	0.0	1.49	0.00	0.00	0.00	0.0	0.00	0.00	1.49	1.49	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.00
Q	0.00	0.0	0.00	0.00	0.00	1.49	0.00	0.0	0.00	0.00	0.00	0.0	1.49	0.00	0.00	0.00	0.00	0.00	0.0	0.00
R	0.00	0.0	0.00	0.00	0.00	0.00	0.0	0.0	0.00	0.00	0.00	0.0	0.00	1.49	0.00	0.0	1.49	0.00	0.0	0.00
S	0.00	0.0	0.00	0.00	0.00	0.00	0.0	0.0	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.00
T	0.00	0.0	0.00	0.00	0.00	0.00	0.0	0.0	0.00	0.00	0.00	0.0	0.00	1.49	0.00	0.0	0.00	0.00	0.0	0.00
V	1.49	0.0	1.49	0.00	0.00	1.49	0.0	0.0	0.00	0.00	0.00	4.48	0.0	0.00	0.00	0.00	0.00	0.00	0.0	0.00
W	0.00	0.0	0.00	0.00	0.00	0.00	0.0	0.0	0.00	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.0	0.00
Y	0.00	0.0	0.00	0.00	0.00	0.00	0.0	0.0	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.0	0.00	1.49	0.0	0.00
1) The residue pair: AA, with an occurrence of 7.463 %																				
2) The residue pair: GL, with an occurrence of 5.97 %																				
3) The residue pair: VM, with an occurrence of 4.478 %																				
4) The residue pair: IG, with an occurrence of 4.478 %																				
5) The residue pair: AV, with an occurrence of 4.478 %																				
6) The residue pair: AI, with an occurrence of 4.478 %																				
7) The residue pair: MG, with an occurrence of 2.985 %																				
8) The residue pair: LG, with an occurrence of 2.985 %																				
9) The residue pair: IP, with an occurrence of 2.985 %																				
10) The residue pair: GI, with an occurrence of 2.985 %																				

Figure 4.2.2: Score 2: $\left[\frac{N_{ij} \times 100}{N-1} \right]$

	A	C	D	E	F	G	H	I	K	L	M	N	P	Q	R	S	T	V	W	Y
A	2.96	0.0	0.00	0.00	0.00	0.00	0.0	2.88	0.0	0.00	0.00	0.0	0.00	0.0	3.85	0.0	0.0	3.85	0.0	0.00
C	0.00	0.0	0.00	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.00	0.0	0.00	0.0	0.00	0.0	0.0	0.00	0.0	0.

For Sequence 3

	A	C	D	E	F	G	H	I	K	L	M	N	P	Q	R	S	T	V	W	Y
A	9.38	0.0	4.00	0.00	4.17	15.38	0.0	4.17	0.00	0.00	0.00	4.76	5.00	0.00	0.00	0.00	4.35	4.17	0.00	0.00
C	0.00	0.0	0.00	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
D	4.00	0.0	0.00	6.67	0.00	3.12	0.0	0.00	6.67	0.00	0.00	0.00	0.00	7.14	7.14	4.17	0.00	5.88	6.67	0.00
E	4.55	0.0	6.67	0.00	7.14	6.90	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	4.76	0.00	0.00	0.00	0.00
F	12.50	0.0	0.00	7.14	0.00	0.00	0.0	6.25	0.00	0.00	0.00	0.00	0.00	0.00	0.00	4.35	0.00	6.25	0.00	6.25
G	7.69	0.0	9.38	0.00	0.00	2.17	0.0	6.45	3.45	9.38	4.17	0.00	0.00	7.14	3.57	2.63	3.33	6.45	0.00	6.45
H	0.00	0.0	0.00	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.88	0.00	0.00	0.00	10.00
I	0.00	0.0	0.00	14.29	0.00	6.45	10.0	0.00	7.14	0.00	0.00	0.00	8.33	0.00	0.00	0.00	0.00	0.00	0.00	0.00
K	0.00	0.0	0.00	0.00	7.14	0.00	12.5	7.14	0.00	0.00	0.00	0.00	10.00	9.09	0.00	4.76	0.00	0.00	0.00	0.00
L	0.00	0.0	0.00	0.00	5.88	6.25	0.0	0.00	13.33	0.00	0.00	7.14	0.00	0.00	0.00	8.33	6.25	0.00	0.00	0.00
M	0.00	0.0	0.00	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	12.50	0.00	0.00	0.00
N	4.76	0.0	7.14	0.00	7.69	3.57	0.0	0.00	0.00	7.14	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
P	0.00	0.0	0.00	0.00	8.33	0.00	0.0	8.33	0.00	0.00	0.00	11.11	0.00	0.00	0.00	0.00	0.00	8.33	0.00	0.00
Q	0.00	0.0	0.00	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.00	10.00	11.11	0.00	0.00	10.00	8.33	0.00	0.00	0.00
R	4.76	0.0	0.00	0.00	0.00	0.00	0.0	7.69	0.00	14.29	0.00	0.00	0.00	0.00	0.00	0.00	0.00	7.69	0.00	0.00
S	3.23	0.0	0.00	0.00	4.35	5.26	0.0	4.35	0.00	12.50	0.00	5.00	0.00	0.00	0.00	3.33	4.55	4.35	9.52	4.35
T	0.00	0.0	0.00	0.00	0.00	10.00	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.33	4.55	0.00	0.00	0.00	13.33
V	4.17	0.0	0.00	0.00	6.25	9.68	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	7.69	4.35	0.00	0.00	0.00	6.25
W	0.00	0.0	20.00	8.33	0.00	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.00	9.09	0.00	0.00	0.00	0.00	8.33	0.00
Y	0.00	0.0	0.00	7.14	0.00	0.00	0.0	0.00	7.14	0.00	0.00	0.00	0.00	0.00	0.00	7.69	8.70	6.67	0.00	14.29
1) The residue pair: WD, with an occurrence of 20.0 %																				
2) The residue pair: AG, with an occurrence of 15.385 %																				
3) The residue pair: YW, with an occurrence of 14.286 %																				
4) The residue pair: RL, with an occurrence of 14.286 %																				
5) The residue pair: IE, with an occurrence of 14.286 %																				
6) The residue pair: TY, with an occurrence of 13.333 %																				
7) The residue pair: LK, with an occurrence of 13.333 %																				
8) The residue pair: SL, with an occurrence of 12.5 %																				
9) The residue pair: MT, with an occurrence of 12.5 %																				
10) The residue pair: KH, with an occurrence of 12.5 %																				

Figure 4.3.1: Score 1: $\left[\frac{N_{ij} \times 100}{N_i + N_j}\right]$

	A	C	D	E	F	G	H	I	K	L	M	N	P	Q	R	S	T	V	W	Y
A	2.00	0.0	0.67	0.00	0.67	4.00	0.00	0.67	0.00	0.00	0.67	0.67	0.00	0.00	0.00	0.67	0.67	0.00	0.00	0.00
C	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
D	0.67	0.0	0.00	0.67	0.00	0.67	0.00	0.00	0.67	0.00	0.67	0.67	0.00	0.67	0.67	0.00	0.67	0.67	0.67	0.00
E	0.67	0.0	0.67	0.00	0.67	1.33	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.67	0.00	0.00	0.00	0.00
F	2.00	0.0	0.00	0.67	0.00	0.00	0.00	0.67	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.67	0.00	0.67	0.00	0.67
G	2.00	0.0	2.00	0.00	0.00	0.67	0.00	1.33	0.67	2.00	0.67	0.00	0.00	1.33	0.67	0.67	0.67	1.33	0.00	1.33
H	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.67	0.00	0.00	0.00	0.67
I	0.00	0.0	0.00	1.33	0.00	1.33	0.67	0.00	0.67	0.00	0.00	0.67	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
K	0.00	0.0	0.00	0.00	0.67	0.00	0.67	0.67	0.00	0.00	0.00	0.67	0.67	0.67	0.00	0.67	0.00	0.00	0.00	0.00
L	0.00	0.0	0.00	0.00	0.67	1.33	0.00	0.00	1.33	0.00	0.00	0.67	0.00	0.00	0.00	1.33	0.67	0.00	0.00	0.00
M	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.67	0.00	0.00	0.00
N	0.67	0.0	0.67	0.00	0.67	0.67	0.00	0.00	0.00	0.67	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
P	0.00	0.0	0.00	0.00	0.67	0.00	0.00	0.67	0.00	0.00	0.00	0.67	0.00	0.00	0.00	0.00	0.00	0.67	0.00	0.00
Q	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.67	0.67	0.00	0.00	1.33	0.67	0.00	0.00	0.00
R	0.67	0.0	0.00	0.00	0.00	0.00	0.00	0.67	0.00	1.33	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.67	0.00	0.00
S	0.67	0.0	0.00	0.00	0.67	1.33	0.00	0.67	0.00	2.00	0.00	0.67	0.00	0.00	0.00	0.67	0.67	0.67	1.33	0.67
T	0.00	0.0	0.00	0.00	0.00	2.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.67	0.67	0.00	0.00	0.00	1.33
V	0.67	0.0	0.00	0.00	0.67	2.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.67	0.67	0.00	0.00	0.00	0.67
W	0.00	0.0	2.00	0.67	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.67	0.00	0.00	0.00	0.00	0.00	0.67	0.00
Y	0.00	0.0	0.00	0.67	0.00	0.00	0.00	0.00	0.67	0.00	0.00	0.00	0.00	0.00	0.67	1.33	0.67	0.00	1.33	0.00
1) The residue pair: AG, with an occurrence of 4.0 %																				
2) The residue pair: WD, with an occurrence of 2.0 %																				
3) The residue pair: VG, with an occurrence of 2.0 %																				
4) The residue pair: TG, with an occurrence of 2.0 %																				
5) The residue pair: SL, with an occurrence of 2.0 %																				
6) The residue pair: GL, with an occurrence of 2.0 %																				
7) The residue pair: GD, with an occurrence of 2.0 %																				
8) The residue pair: GA, with an occurrence of 2.0 %																				
9) The residue pair: FA, with an occurrence of 2.0 %																				
10) The residue pair: AA, with an occurrence of 2.0 %																				

Figure 4.3.2: Score 2: $\left[\frac{N_{ij} \times 100}{N-1}\right]$

	A	C	D	E	F	G	H	I	K	L	M	N	P	Q	R	S	T	V	W	Y
A	1.17	0.0	0.69	0.00	0.78	1.63	0.00	0.78	0.00	0.00	0.00	1.25	1.56	0.00	0.00	0.00	0.89	0.78	0.00	0.00
C	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
D	0.69	0.00	0.00	1.85	0.00	0.48	0.00	0.00	1.85	0.00	0.00	0.00	0.00	2.22	2.22	0.74	0.00	1.39	1.85	0.00
E	1.04	0.0	1.85	0.00	2.08	1.45	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.11	0.00	0.00	0.00	0.00
F	2.34	0.00	0.00	2.08	0.00	0.00	0.00	1.56	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.83	0.00	1.56	0.00	1.56
G	0.82	0.0	1.45	0.00	0.00	0.19	0.00	1.09	0.72	1.45	4.35	0.00	0.00	1.74	0.87	0.29	0.62	1.09	0.00	1.09
H	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.33	0.00	0.00	0.00	6.25
I	0.00	0.00	0.00	4.17	0.00	1.09	6.25	0.00	2.08	0.00	0.00	0.00	3.12	0.00	0.00	0.00	0.00	0.00	0.00	0.00
K	0.00	0.00	0.00	0.00	2.08	0.00	8.33	2.08	0.00	0.00	0.00	0.00	4.17	3.33	0.00	1.11	0.00	0.00	0.00	0.00
L	0.00	0.00	0.00	0.00	1.39	0.97	0.00	0.00	3.70	0.00	0.00	2.22	0.00	0.00	0.00	1.48	1.59	0.00	0.00	0.00
M	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	14.29	0.00	0.00	0.00
N	1.25	0.0	2.22	0.00	2.50	0.87	0.00	0.00	0.00	2.22	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
P	0.00	0.00	0.00	0.00	3.12	0.00	0.00	3.12	0.00	0.00	0.00	5.00	0.00	0.00	0.00	0.00	0.00	3.12	0.00	0.00
Q	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	4.00	5.00	0.00	0.00	2.67	2.86	0.00	0.00	0.00
R	1.25	0.00	0.00	0.00	0.00	0.00	0.00	2.50	0.00	4.44	0.00	0.00	0.00	0.00	0.00	0.00	0.00	2.50	0.00	0.00
S	0.42	0.00	0.00	0.00	0.83	0.58	0.00	0.83	0.00	2.22	0.00	1.33	0.00	0.00	0.00	0.44	0.95	0.83	2.22	0.83
T	0.00	0.00	0.00	0.00	0.00	1.86	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	2.86	0.95	0.00	0.00	0.00	3.57
V	0.78	0.00	0.00	0.00	1.56	1.63	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	2.50	0.83	0.00	0.00	0.00	1.56
W	0.00	0.0	5.56	2.78	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.33	0.00	0.00	0.00	0.00	2.78	0.00
Y	0.00	0.00	0.00	2.08	0.00	0.00	0.00	0.00	2.08	0.00	0.00	0.00	0.00	0.00	2.50	1.67	1.79	0.00	4.17	0.00
1	The residue pair: MT, with an occurrence of 14.286 %																			
2	The residue pair: KH, with an occurrence of 8.333 %																			
3	The residue pair: IH, with an occurrence of 6.25 %																			
4	The residue pair: HY, with an occurrence of 6.25 %																			
5	The residue pair: WD, with an occurrence of 5.556 %																			
6	The residue pair: QP, with an occurrence of 5.0 %																			
7	The residue pair: PN, with an occurrence of 5.0 %																			
8	The residue pair: RL, with an occurrence of 4.444 %																			
9	The residue pair: GM, with an occurrence of 4.348 %																			
10	The residue pair: YW, with an occurrence of 4.167 %																			

Question 5

The code to calculate average hydrophobicity (Hgm), Helical contact area (Ca) and Total non-bonded energy (Et) is given below:

```
Et = "1.90  1.52  2.04  1.54  1.86  1.90  1.76  1.95  1.37  1.97  1.96  
1.56  1.70  1.52  1.48  1.75  1.77  1.98  1.87  1.69"  
  
Ca = "20.00  26.00  25.00  33.00  46.00  13.00  37.00  39.00  46.00  35.00  
43.00  28.00  22.00  36.00  55.00  20.00  28.00  33.00  61.00  46.00"  
  
Hgm = "13.85  11.61  15.37  11.38  13.93  13.34  13.82  15.28  11.58  14.13  
13.86  13.02  12.35  12.61  13.10  13.39  12.70  14.56  15.48  13.88"  
  
AAs = "ADCEFGHIKLMNPQRSTVWY"  
  
Et_temp = Et.split(" ")  
Et = {}  
  
Ca_temp = Ca.split(" ")  
Ca = {}  
  
Hgm_temp = Hgm.split(" ")  
Hgm = {}  
  
for n in range(len(AAs)):  
    Et[AAs[n]] = float(Et_temp[n])  
    Ca[AAs[n]] = float(Ca_temp[n])  
    Hgm[AAs[n]] = float(Hgm_temp[n])  
  
def Scoring(seq):  
    score_Hgm, score_Ca, score_Et = 0, 0, 0  
    for AA in seq:  
        score_Hgm += Hgm[AA]  
        score_Ca += Ca[AA]  
        score_Et += Et[AA]  
  
    avg_Hgm = score_Hgm/len(seq)  
  
    print(f"Average hydrophobicity for the given sequence is: {round(avg_Hgm,2)}")  
    print(f"Helical contact area for the given sequence is: {score_Ca}")  
    print(f"Total non-bonded energy for the given sequence is:{round(score_Et,3)}")  
    print("=*100)  
  
Scoring(seq1)  
Scoring(seq2)  
Scoring(seq3)
```

The output for the above code is shown below:

```
Average hydrophobicity for the given sequence is: 13.35
Helical contact area for the given sequence is: 2156.0
Total non-bonded energy for the given sequence is: 117.74
```

```
=====
Average hydrophobicity for the given sequence is: 13.77
Helical contact area for the given sequence is: 2067.0
Total non-bonded energy for the given sequence is: 126.66
```

```
=====
Average hydrophobicity for the given sequence is: 13.42
Helical contact area for the given sequence is: 4616.0
Total non-bonded energy for the given sequence is: 267.75
=====
```

- All sequences seem to be hydrophobic (with a nearly equal average hydrophobicity value). The order being Sequence 2 > Sequence 3 > Sequence 1 (which we predicted in Question 1).
- Sequence 3 has a very high total contact area (nearly 2 times the other sequences' total contact area).
- Sequence 3 also has a very high total non-bonded energy as well (again nearly double the other sequences' total non-bonded energy).

Link to the google drive containing the python code used in this assignment: https://drive.google.com/drive/u/0/folders/1S_NK7kwsRdEi4wHEYMApJMxccr6gRsU

Annotations have been omitted from the submitted PDF for the sake of brevity. The fully annotated version of the code is available in the code folder within the attached Google Drive.
